SOLUTION - FINAL EXAM

FALL - 2024

CALCULUS AND ANALYTICAL GEOMETRY

Q1 (a) (i) $f(-1) = 3$ (from graph)

(ii) $g(2) = 5 - 2(2) = 5 - 4 = 1$

and $f(g(2)) = f(1) = 7$ (from graph)

(iii) $\lim_{x \to 1^+} f(x) = 2$ (right-hand limit at $x = 1$, visible in graph)

$\lim_{x \to 1^-} f(x) = 7$ (left-hand limit at $x = 1$,

(b) $y(t) = \frac{e^{-2t} - 1}{2}$, $t \geq 0$

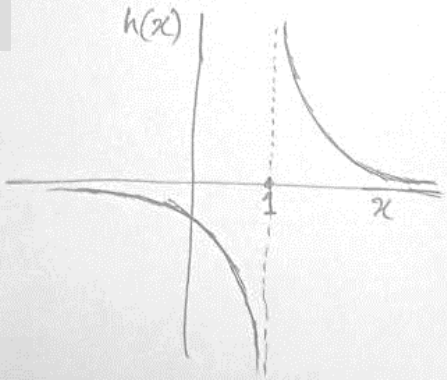
Domain is $t \in [0, \infty)$ as given above.

At $t = 0$, $y(0) = \frac{e^{-0} - 1}{2} = \frac{1-1}{2} = 0$

At $t \to \infty$, $y(\infty) = \frac{e^{-\infty} - 1}{2} = \frac{0-1}{2} = -\frac{1}{2}$

Hence, range: (-1/2, 0]

(c) $h(x) = \frac{1}{x-1}$



Q1(d)

$$\lim_{y\to 0} \frac{5y^3+8y^2}{3y^4-16y^2} = \lim_{y\to 0} \frac{y^2(5y+8)}{y^2(3y^2-16)}$$

$$= \frac{5\times 0+8}{3\times 0^2-16} = \frac{8}{-16} = -\frac{1}{2}$$

Q2(a) $x^2\cos^2 y - \sin y = 0 \quad (0,\pi)$

Differentiate both sides w.r.t. x

$$\frac{d}{dx}(x^2\cos^2 y) - \frac{d}{dx}(\sin y) = 0$$

$$2x\cos^2 y + x^2 2\cos y(-\sin y)\frac{dy}{dx} - \cos y\frac{dy}{dx} = 0$$

$$\frac{dy}{dx}\left[-2x^2\cos y\sin y - \cos y\right] = -2x\cos^2 y$$

$$\frac{dy}{dx} = \frac{-2x\cos^2 y}{-2x^2\cos y\sin y - \cos y}$$

At $(x,y)=(0,\pi)$, $\frac{dy}{dx} = \frac{-2(0)\cos^2(\pi)}{-2(0)^2\cos(\pi)\sin(\pi)-\cos\pi} = 0$

i.e. slope of tanget at $(0,\pi)$ is $m=0$

Hence, ~~line of~~ the equation of tangent is

$$\boxed{y=\pi}$$

Q 2 (b)

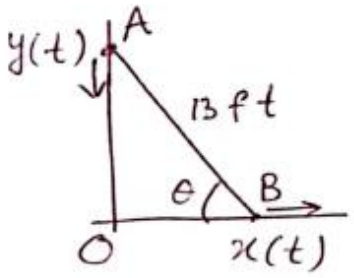
(i) At $t = 0, 3.2, 5.1$ seconds, $v = 0$ as obvious from the graph, i.e. the particle is not moving at $t = 0, 3.2, 5.1$ seconds.

(ii) At $t = 1.9$ sec, 4 sec. and 6 sec., the tangents to the velocity–time graph are horizontal lines, i.e. $\left.\frac{dv(t)}{dt}\right|_{t=1.9,4,6} = 0$

i.e. acceleration is zero at $t = 1.9, 4, 6$ seconds.

(iii) The particle changes direction at $t = 3.2$ sec. and $t = 5.1$ sec.

Q 2 (c)



When $x(t) = 12$-ft,

then $\frac{dx(t)}{dt} = 5$ ft/s

(i)

In $\triangle AOB$,

$(\overline{AO})^2 + (\overline{OB})^2 = (\overline{AB})^2$

$y^2(t) + x^2(t) = 13^2 = 169$

$$2y(t)\frac{dy(t)}{dt} + 2x(t)\frac{dx(t)}{dt} = \frac{d}{dt}(169)$$

$$2y(t)\frac{dy(t)}{dt} = 0 - 2x(t)\frac{dx(t)}{dt}$$

When $x(t) = 12$, then $y^2(t) + 12^2 = 13^2$

$y^2(t) = 169 - 144 = 25 \Rightarrow y(t) = 5$ ft

$$\Rightarrow 2(5)\frac{dy(t)}{dt} = -2(12)(5)$$

$$\Rightarrow \boxed{\frac{dy(t)}{dt} = -12 \text{ ft/s}}$$

Q2 (c) (ii)

Area of $\Delta AOB = A = \frac{1}{2} x(t)\, y(t)$

$$\frac{dA}{dt} = \frac{1}{2} x(t) \frac{dy(t)}{dt} + \frac{1}{2} y(t) \frac{dx(t)}{dt}$$

$$= \frac{1}{2}(12)(-12) + \frac{1}{2}(5)(5) = \frac{-144}{2} + \frac{25}{2}$$

$$\boxed{\frac{dA}{dt} = -\frac{119}{2} \text{ ft}^2/\text{s}}$$

-ve sign indicates that area is decreasing

Q3

(a) (i) $\int \left(1-\cos\frac{t}{2}\right)^2 \sin\frac{t}{2}\,dt$,

Using substitution,

$u = 1-\cos\frac{t}{2} \Rightarrow \frac{du}{dt} = 0+\left(\sin\frac{t}{2}\right)\frac{1}{2}$

$\Rightarrow \sin\frac{t}{2}\,dt = 2\,du$

Thus, $\int \left(1-\cos\frac{t}{2}\right)^2 \sin\frac{t}{2}\,dt = \int u^2\,2\,du$

$= \frac{2u^3}{3} + C = \frac{2}{3}\left[1-\cos\frac{t}{2}\right]^3 + C$

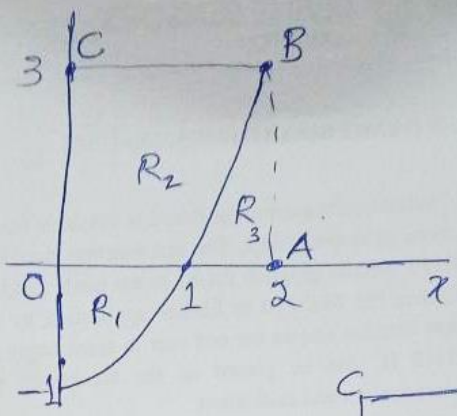
(ii) $\int_0^2 \left(\frac{t^2}{4} - 7t + 5\right)dt = \left[\frac{t^3}{4\times 3} - 7\frac{t^2}{2} + 5t\right]_0^2$

$= \frac{2^3-0^3}{12} - \frac{7}{2}(2^2-0^2) + 5(2-0)$

$= \frac{8}{12} - \frac{7}{2}\times\frac{2}{4} + 10 = \frac{8}{12} - 14 + 10$

$= \frac{2}{3} - 4 = \frac{2-12}{3} = -\frac{10}{3}$

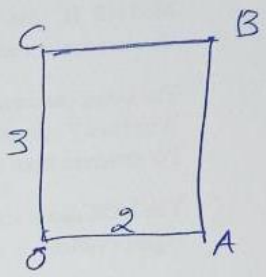
Q3 (b)



Area of region R_1 is $A_1 = \left| \int_0^1 (x^2 - 1)\, dx \right|$

$= \left| \left[\frac{x^3}{3} - x \right]_0^1 \right|$

$= \left| \frac{1}{3} - 1 \right| = \left| \frac{-2}{3} \right| = \frac{2}{3}$



Area of region $R_3 = A_3 = \int_1^2 (x^2 - 1)\, dx = \left[\frac{x^3}{3} - x \right]_1^2$

$= \frac{2^3 - 1^3}{3} - (2 - 1) = \frac{8-1}{3} - 1 = \frac{7}{3} - 1 =$

Area of rectangle $OABC = 2 \times 3 = 6$

Area of $R_2 = A_2$ = area of rect. $OABC - A_3$

$\Rightarrow A_2 = 6 - \frac{4}{3} = \frac{18 - 4}{3} = \frac{14}{3}$

Hence, area of shaded region $= A_1 + A_2$

$= \frac{2}{3} + \frac{14}{3} = \frac{16}{3}$